Linux Fundamentals - Day 3

⭐ Introduction to Linux

✔ Linux is an open-source operating system.

✔ Linux is widely used in servers, cloud computing, networking, and cyber security.

✔ Linux is known for stability, security, and flexibility.

✔ Many cyber security professionals use Linux daily.

⭐ History of Linux

✔ Linux was created by Linus Torvalds in 1991.

✔ Linux started as a personal project and later became one of the most popular operating systems.

✔ Today Linux powers millions of devices worldwide.

⭐ Open Source Concept

✔ Open source means the source code is publicly available.

✔ Developers can study, modify, and improve the code.

✔ Community contributions help Linux evolve rapidly.

Benefits

• Free to use

• Highly customizable

• Large community support

• Strong security updates

⭐ Linux Kernel

✔ The kernel is the core component of Linux.

✔ It acts as a bridge between software and hardware.

✔ It manages:

• CPU

• Memory

• Storage

• Devices

• Processes

Simple Architecture

Applications

↓

Shell

↓

Kernel

↓

Hardware

⭐ Linux Distributions

✔ A Linux Distribution (Distro) is a complete operating system built around the Linux kernel.

Popular Linux Distributions

✔ Ubuntu

• Beginner friendly

• Popular for learning

✔ Kali Linux

• Designed for cyber security

• Includes penetration testing tools

✔ Debian

• Stable and reliable

• Popular for servers

✔ Red Hat Enterprise Linux (RHEL)

• Enterprise environments

• Commercial support

✔ CentOS

• Server-oriented distribution

⭐ Linux Architecture

Components

✔ Hardware

• CPU

• RAM

• Storage

• Network Devices

✔ Kernel

• Controls hardware

• Manages resources

✔ Shell

• User interface for commands

• Communicates with kernel

✔ Applications

• Browsers

• Editors

• Security tools

Architecture Diagram

User

↓

Applications

↓

Shell

↓

Kernel

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Hardware

⭐ Linux File System

Root Directory

✔ Everything starts from:

/

Important Directories

✔ /home

• Stores normal user files

• Example:

/home/jaswanth

✔ /root

• Home directory of administrator

✔ /etc

• Configuration files

• System settings

✔ /usr

• Installed applications and software

✔ /var

• Logs and variable data

• Important for cyber security investigations

✔ /tmp

• Temporary files

✔ /bin

• Essential user commands

✔ /sbin

• Administrative commands

File System Diagram

/

├── home

├── root

├── etc

├── usr

├── var

├── tmp

├── bin

└── sbin

⭐ Linux Boot Process

Step 1

✔ BIOS / UEFI starts hardware initialization.

Step 2

✔ Bootloader (GRUB) loads.

Step 3

✔ Linux Kernel loads into memory.

Step 4

✔ Systemd starts services and processes.

Boot Flow

Power On

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BIOS / UEFI

↓

GRUB

↓

Kernel

↓

Systemd

↓

Login Screen

⭐ Linux Shell

✔ Shell allows users to interact with Linux.

✔ Bash (Bourne Again Shell) is the most commonly used shell.

Examples

• ls

• pwd

• cd

• mkdir

Benefits

✔ Fast

✔ Efficient

✔ Automation Friendly

✔ Essential for Cyber Security

⭐ Linux vs Windows

Linux

✔ Open Source

✔ Free

✔ Highly Customizable

✔ Popular in Servers

Windows

✔ Closed Source

✔ Commercial Product

✔ GUI Focused

✔ Popular on Personal Computers

⭐ Kali Linux

✔ Kali Linux is a cyber security focused Linux distribution.

✔ Built on Debian.

✔ Contains hundreds of security tools.

Popular Tools

✔ Nmap

• Network scanning

✔ Wireshark

• Packet analysis

✔ Metasploit

• Security testing

✔ Burp Suite

• Web security testing

⭐ Linux in Cyber Security

SOC Analyst

✔ Monitor logs

✔ Investigate incidents

✔ Analyze alerts

Security Engineer

✔ Automate security tasks

✔ Manage servers

✔ Deploy security tools

Penetration Tester

✔ Vulnerability assessment

✔ Security testing

✔ Reconnaissance

Incident Responder

✔ Analyze compromised systems

✔ Collect evidence

✔ Investigate attacks

⭐ Key Learning Outcomes

✔ Understand Linux Fundamentals

✔ Understand Open Source Concept

✔ Understand Linux Kernel

✔ Learn Linux Distributions

✔ Learn Linux Architecture

✔ Learn Linux File System

✔ Learn Linux Boot Process

✔ Understand Kali Linux

✔ Understand Linux Usage in Cyber Security

⭐ Quick Revision

✔ Linux = Open Source Operating System

✔ Kernel = Core of Linux

✔ Ubuntu = Beginner Friendly

✔ Kali Linux = Cyber Security

✔ /home = User Files

✔ /root = Administrator Files

✔ /etc = Configuration Files

✔ /var = Log Files

✔ Bash = Popular Linux Shell

✔ Linux is widely used in Cyber Security and Servers